HOUSEHOLD CIRCUITS

Time: 1hr **Marks:40**

SECTION I

[Attempt *all* questions from this section]

Question 1: **[5]**

[i] The potential difference between an earth wire and neutral wire is .....

[a] very low

[b] very high

[c] zero

[d] both a and c

[ii] Which of the following is the correct colour code of wiring

[a] Live: green, Earth: yellow, Neutral: red

[b] Live: brown, Earth: blue, Neutral: red

[c] Live: brown, Earth: greenish yellow, Neutral: blue

[d] Live: red, Earth: blue, Neutral: yellow

[iii] Which of the following statements is true?

[a] as the radius of filament wire increases the temperature of fuse wire increases more than its surroundings.

[b] as the radius of filament wire increases the temperature of fuse wire drops below its surroundings.

[c] as current flowing through the filament wire increases the temperature of fuse wire rises above its surroundings.

[d] both b and c

[iv] Ring system is an example of……

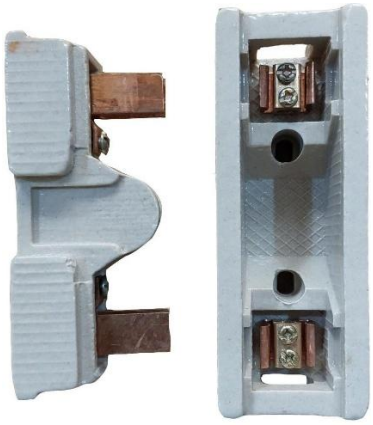
[a] series type of circuit connection

[b] parallel type of circuit connection

[c] both series and parallel type of circuit connection

[d] none of the above.

[v] Which of the two should be preferred in household circuits



[i]



[ii]

[a] [i]

[b] [ii]

[c] both [i] and [ii]

[d] none of the above

Question 2:

Fill in the blanks and rewrite the complete sentence. [5]

[i] The devise which makes or breaks an electric circuit is …. [bulb / plug key / rheostat]

[ii] Earthing is done to an appliance as earth is considered to be at …….. [infinite / zero] potential.

[iii] Composition of fuse is an alloy of ……. [mercury and tin / aluminum and copper / lead and tin].

[iv] Earth pin is thicker than live and neutral as it offers ……..

[maximum / minimum] resistance.

[v] An open circuit is caused due to….. [live wire not connected to neutral / plug key being closed very tight / fuse melting].

Question 3:

[a] Under what circumstances does one get an electric shock from a gadget. [any two] [2]

[b] Which properties of copper make it an unsuitable material for it to be used as a fuse wire. [2]

[c] Which of the two cables, one rated 5A and other 20A would be thicker? Justify your inference. [2]

[d] What is the purpose of fuse in an electric circuit? It connected to which wire in the household circuit? [2]

[e] State two characteristics of high-tension wires used in transmission lines from substation to end users. [2]

SECTION II

[Attempt *any two* questions out of three]

Question 4:

[a] [i] Name the protective device used to protect the wiring system from overloading or short circuiting.

[ii] On what effect of electric current does the above device work. [4]

[b] The wires used to carry current from mains to the appliance are made of copper while the filament of a bulb is made of tungsten. Justify this statement with suitable reason. **[3]**

[c] Explain the term short circuiting used in household circuits **[3]**

Question 5:

[a] [i] Which two wires in the wiring system are at the same potential for all practical purposes?

[ii] What is the importance of earthing in an electric circuit? **[4]**

[b] State the old colour code of wiring. **[3]**

[c] Why is ring system preferred in household wiring? **[3]**

Question 6:

[a]

[i] What do you understand by rating of an electrical appliance?

[ii] What is meant by the statement "the rating of an electric bulb is 100W-230V"? **[4]**

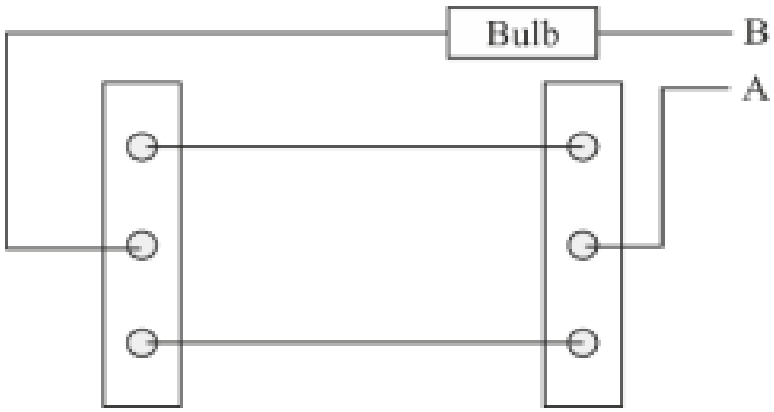
[b] How does the heat produced by the passage of current in a metallic wire depend on:

[i] resistance of wire

[ii] current flowing in the wire

[iii] time for which current flows through the wire. **[3]**

[c] The given diagram shows a dual control switch circuit which is connected to the bulb. **[3]**



[i] Copy the diagram and complete it so that the bulb is switched ON when used by left hand side switch.

[ii] Out of the two A and B which is live wire and which wire is neutral.

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